

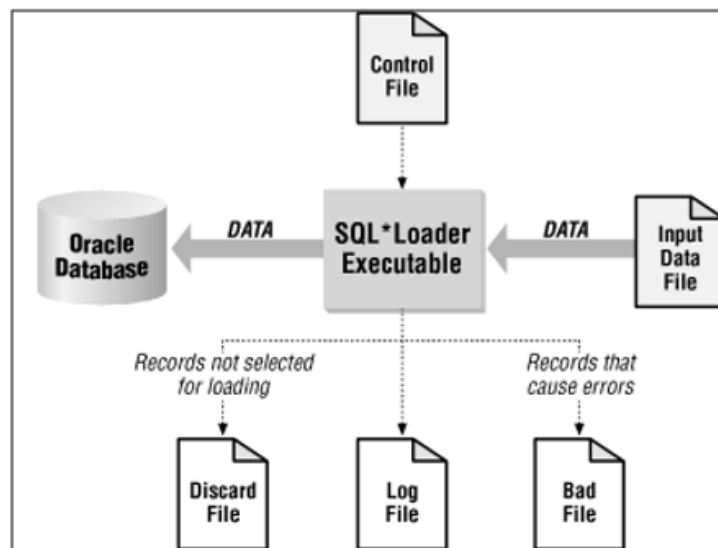
Query Optimization: SQL Loader, Explain Plan, and Autotrace

This second lab introduces fundamental concepts and practical applications of *DB optimization*. We will explore essential techniques to improve the performance of *relational databases*. The session begins with *SQL Loader*, a versatile tool designed for efficiently loading large datasets into Oracle databases. Additionally, we will examine the *Explain Plan* and *Autotrace* utilities, which offer valuable insights into the execution plans of *SQL queries*.

① Useful Notions & Syntaxes

(a) SQL Loader

- i) It is a specialized loading utility for *Oracle databases*, enabling the initialization of tables with data from a text file.



ii) Key features

- Loads external text files into Oracle (such as .CSV, etc) in a single load session, which can enhance efficiency when dealing with large datasets;
- Powerful data analyzing engine imposes minimal restrictions on data file formats, allowing it to handle a wide variety of data structures (',' or '...');
- *Multiple Data File Handling*, it can load data from multiple data files;
- Uses of SQL functions;
- Unique key generation, i.e., it can automatically generate sequence numbers;
- *Multiple Loading Methods*, it can offer both loading methods *conventional path* (i.e., Loads data one row at a time using standard *INSERT* statements) and *direct path* (i.e.,

Data is written directly into the data blocks, Much faster). Each method has its advantages and use cases, depending on factors such as performance, speed, constraints, and logging requirements, allowing users to choose based on their needs. For our **Lab** we will employ the optimized *direct mode*;

-*Comprehensive Error Reporting*, it generates detailed error reports, aiding in troubleshooting and ensuring data integrity, i.e., *Log Management, Error Handling, and Recovery Capabilities*.

iii) **General Principle**

- The destination table(s) should be created in the target schema, ensuring that the table names match exactly.
- Input and output formats are defined in a parameter file known as the control file, which can be created using a text editor (.ctl). A log file will be generated to record the results of the loading process (.log). If any errors occur, the rejected records will be stored in a (.bad) file, which can be reprocessed if needed.

iv) **Use**

- Create the control file as *controleFile.ctl* (from a text file, separated files *recommended*), as:

```
LOAD DATA
INFILE 'datafile.csv'---the path
INTO TABLE my_table
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
TRAILING NULLCOLS
(
    column1,
    column2,
    column3
)
```

- Note that it is important to know that *SQL Loader* is not going to create the tables if they are not already created. The following table contains some of the useful command line options:

- To start the loading, use the Command System prompt:

```
sqlldr userid=username/password control=loader.ctl log=loader.log bad=badfile.bad direct=true
```

- **Loading Modes**

- **Insert**. Inserts data into an empty table (default option);
- **Append**. Inserts data after the existing data;
- **Replace**. Inserts data by replacing the existing data.

OPTION	DESCRIPTION	DEFAULT
userid	username and password	
control	name of the loader control file	
data	name of the input data file	
log	name of the trace file	
bad	name of the rejected records file	
skip <i>n</i>	number of logical records to skip	0
load <i>n</i>	number to load	all
errors <i>n</i>	maximum number of allowed errors	50
rows <i>n</i>	number of array rows to use for input	64
bindsize	size of the preceding array in bytes	OS-dependent
silent	do not display messages during execution	
direct	use direct path (direct path)	false
discard	file for intentionally unloaded records (conditional skip)	
discardmax	maximum number of these intentionally unloaded records	all

SQLLoader (.ctl file) and SQLPlus (CREATE TABLE statements) use different type definitions, so you need to adjust the types when writing your control file for SQL*Loader, as follows:

- CHAR(length) → For VARCHAR2 and CHAR
- INTEGER EXTERNAL → For INTEGER
- DECIMAL EXTERNAL → For NUMBER
- DATE "YYYY-MM-DD" → For DATE

(b) **Explain Plan**

i) **EXPLAIN PLAN**

The **EXPLAIN PLAN** statement is used to generate the execution plan for a *SQL query in Oracle Database*. It does not execute the query itself; rather, it invokes the *Oracle optimizer to analyze* the query and provides a detailed execution plan. The basic syntax is as follows:

```
EXPLAIN PLAN [SET STATEMENT_ID = 'statement_id'] FOR
SQL_Statement;
```

Where:

1. *EXPLAIN PLAN*: This is the primary statement used to generate the execution plan for an SQL query.
2. *SETSTATEMENT_ID*: This optional parameter assigns a unique identifier (*statement_id*) to the execution plan. It can be useful for referencing or distinguishing between different plans in the *PLAN_TABLE*.
3. *SQL_Statement*: This represents the query (usually a DML statement) for which the execution plan is generated.

- ii) *PLAN_TABLE*: This is a system table in Oracle Database where execution plans generated by **EXPLAIN PLAN** are stored. It provides a structured format for viewing the details of the execution plan, including information such as the order of operations, access methods, and join methods. The basic syntax is as follows:

```
-- Display the execution plan from the PLAN_TABLE
SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY('statement_id'));
```

(c) AUTOTRACE

- i) **AUTOTRACE**. This is another feature in SQLPlus that allows the display of the execution plan and statistics for a SQL statement after it has been executed. It presents a quick assessment of the performance characteristics of a query without specifically running the *EXPLAIN PLAN* statement and then executing the query separately.

It is important to prepare the user to work with *AUTOTRACE*. Therefore, it is needed to ensure that the user has the necessary privileges and settings. Here are the steps you can follow:

1. The user should have the necessary privileges to run the queries they want to analyze using *AUTOTRACE*;
2. Grant the necessary *roles* to the user. The *SELECT_CATALOG_ROLE* role, for example, includes privileges to access the data dictionary views needed. As following:

```
GRANT SELECT_CATALOG_ROLE TO userName;
```

3. Enable Server Output to be able to see the results of the *AUTOTRACE* command.

```
SET SERVEROUTPUT ON;
```

4. *Enable AUTOTRACE*. The user can enable *AUTOTRACE* for their session by using the following command:

```
SET AUTOTRACE ON;
SQL STATEMENT;
```

5. *Optional Settings*. The output can be customized by using additional settings, to show both the execution plan *EXPLAIN* and the *statistics* after the query is executed, For example:

```
SET AUTOTRACE ON EXPLAIN STATISTICS;
SQL STATEMENT;
```

6. Disable *AUTOTRACE*, as:

```
SET AUTOTRACE OFF;
```

Important Remark

SET TIMING ON command in SQLPlus is used to enable the display of time for the execution of a SQL statement. Once it is turned on, SQLPlus will display the time it takes for a query or command to be executed.

② Lab Work

- i) Use SQL Loader to upload the (.CSV) file into imijdbManager DB Tables.
- ii) **We have the following set of queries: Display the execution plan, The elapsed time, and Explain the results. What do you notice?**
 - **Use the Quereies from Lab1 As well.**
 - Retrieve episodes with their season and episode numbers for a specific show, e.g. idShow = 989125, and idShow= 47768;
 - Retrieve the average rating and number of votes for a specific show, e.g. idShow = 621278, and idShow= 10140334 ;
 - Retrieve staff members who were involved as principals in a specific show, e.g. idShow = 23246, and idShow= 4324;
 - Calculate the number of shows with aveRating $\geq$ 6.1;
 - Calculate the number of shows with showType equal to " tvSeries", and " tvEpisode";
 - Retrieve the names of staff members and the titles of shows they have participated in as principals, along with the corresponding categories.
 - Find the average rating for each staff member based on the shows they have participated in.
 - Retrieve the top 5 TV series and short with the highest average ratings.
 - Retrieve all shows, along with their average ratings if available. Explain what you notice.

③ Homework

By the end of each Lab, a report should be done, that contains all the details of the elaborate work during the Lab session. The report is going to be submitted later on. **You can work in groups of four maximum.**